EXPERIMENT NO: 6

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Subject Name: System Design
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Aim

To design a highly scalable video streaming platform similar to YouTube that supports heavy video uploading, adaptive bitrate streaming, and real-time social interactions with high availability and low latency.

Objectives

- Understand the architecture of a massive-scale media streaming system.
- Identify functional requirements such as direct-to-cloud uploads and personalized feeds.
- Identify non-functional requirements including CDN caching, high throughput, and latency constraints.
- Analyze the decoupling of synchronous API requests using event-driven stream processing.
- Design REST APIs for video management, search, and user interactions.

Procedure

1. Studied real-world video streaming architectures and content delivery networks.
2. Identified core entities such as Users, Videos, Interactions, and View Histories.
3. Analyzed the Pre-Signed URL pattern to bypass API Gateway bottlenecks for large file uploads.
4. Collected functional and non-functional requirements for media delivery.
5. Designed APIs for video processing, metadata retrieval, and social engagement.
6. Evaluated streaming protocols (HLS/DASH) and separated thumbnail storage for optimization.
7. Analyzed stream processing (Apache Flink/Kafka) for real-time, high-volume view aggregation.

Functional Requirements

1. System should allow users to register, log in, and manage accounts.
2. System should support uploading large raw video files efficiently.
3. System should stream video smoothly using Adaptive Bitrate Streaming based on network conditions.
4. System should support real-time social interactions (likes, dislikes, comments).
5. System should allow users to search for videos by title or metadata.

6. System should generate a personalized home page feed based on watch history.

Core Entities of the System

- Users
- Videos
- Interactions (Likes/Dislikes)
- Comments
- ViewHistory

API Design

1. Authentication API

A. **Register**: POST /api/v1/auth/register
B. **Login**: POST /api/v1/auth/login

2. Video & Feed API

A. **Initialize Upload**: POST /api/v1/videos/upload/init (Returns S3 Pre-Signed URL)
B. **Get Video Metadata**: GET /api/v1/videos/{video_id}
C. **Get Personalized Feed**: GET /api/v1/feed (Pagination Supported)
D. **Search Videos**: GET /api/v1/search?q={query}

3. Interaction API

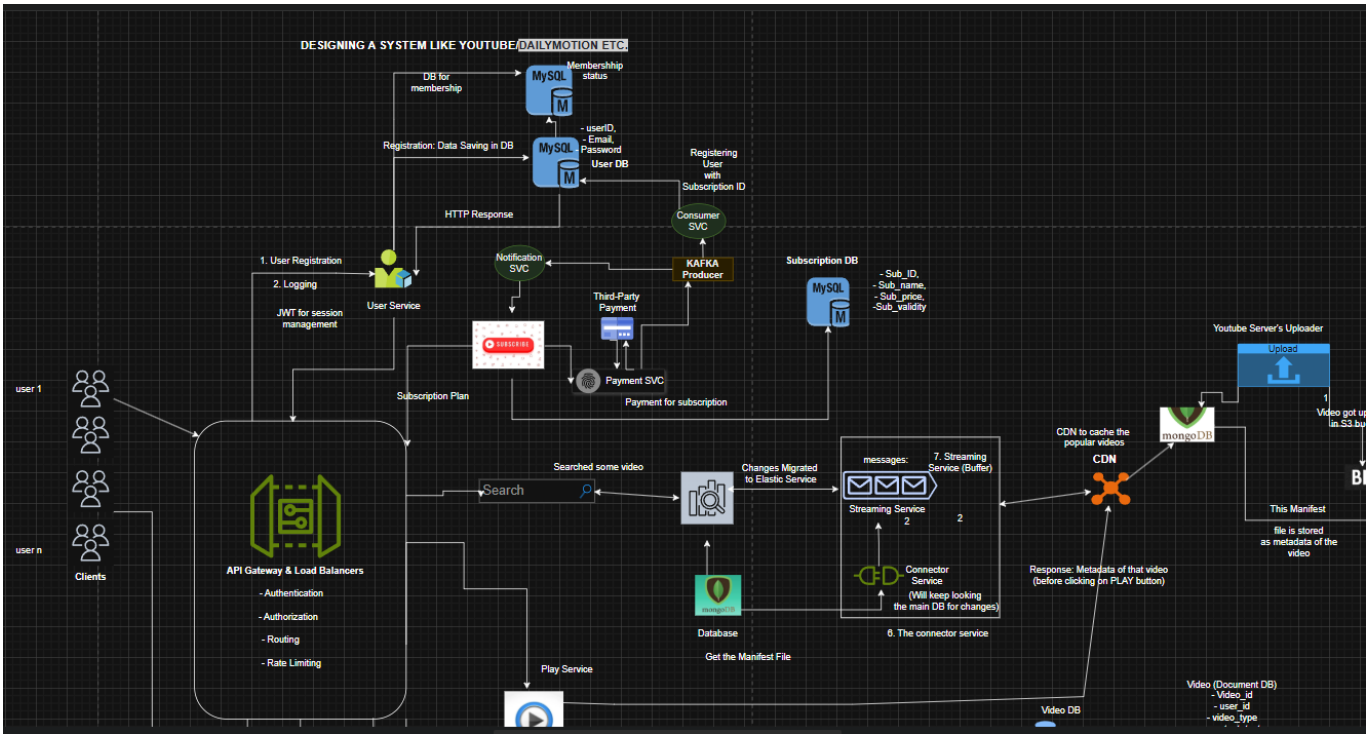
A. **Like Video**: POST /api/v1/videos/{video_id}/like
B. **Add Comment**: POST /api/v1/videos/{video_id}/comments
C. **Get Comments**: GET /api/v1/videos/{video_id}/comments

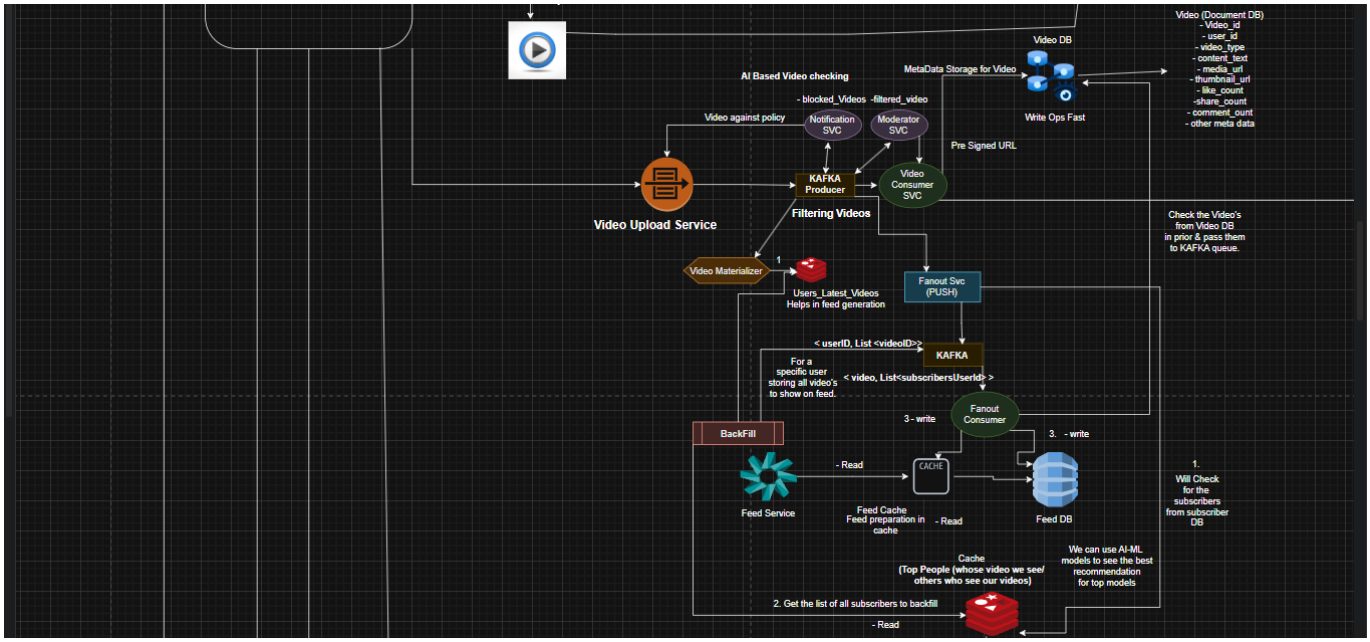
Non-Functional Requirements

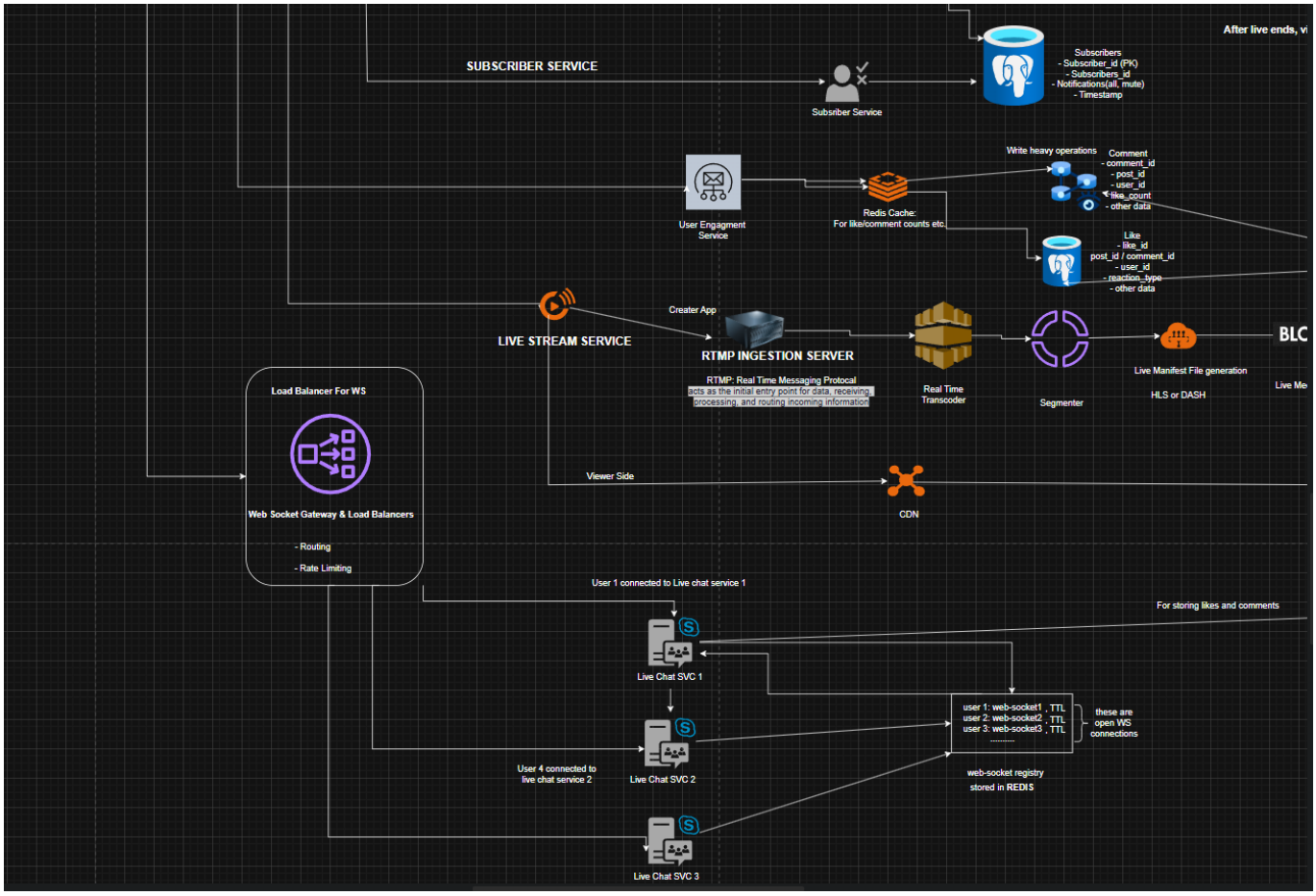
1. System must handle extremely high throughput for concurrent uploads and millions of simultaneous views.
2. High Availability must be prioritized, ensuring no single point of failure.
3. System should follow an Eventual Consistency model for social metrics (view counts, likes) to optimize database write loads.
4. Video delivery latency should be minimized via Edge Caching and a Global CDN.
5. Transcoding pipeline must be horizontally scalable to handle viral upload spikes.

High Level Design (HLD)

The system consists of Client Applications, an API Gateway, and specialized microservices (Auth, Upload, Video Metadata, Feed, and Interactions). Heavy uploads bypass the backend via Pre-Signed URLs directly to Object Storage. Asynchronous transcoding pipelines process videos into HLS/DASH segments, which are delivered globally via a CDN. High-volume interactions are managed using Redis (Likes), NoSQL clusters (Comments), and a Kafka-to-Apache Flink stream processing pipeline for accurate view aggregation.


DESIGNING A SYSTEM LIKE YOUTUBE/DAILYMOTION ETC.
DB for membership
MySQL
Membership status
Registration: Data Saving in DB
MySQL
- userID,
- Email,
- Password
User DB
Registering User with Subscription ID
HTTP Response
Consumer SVC
1. User Registration
2. Logging
User Service
JWT for session management
Notification SVC
KAFKA Producer
Subscription DB
MySQL
- Sub_ID,
- Sub_name,
- Sub_price,
-Sub_validity
Third-Party Payment
Payment SVC
Payment for subscription
Subscription Plan
user 1
user n
Clients
API Gateway & Load Balancers
- Authentication
- Authorization
- Routing
- Rate Limiting
Searched some video
Search
Changes Migrated to Elastic Service
messages:
7. Streaming Service (Buffer)
Streaming Service
2
2
Connector Service
(Will keep looking the main DB for changes)
6. The connector service
Database
Get the Manifest File
Play Service
CDN to cache the popular videos
CDN
Youtube Server's Uploader
Upload
mongoDB
This Manifest
file is stored as metadata of the video
Response: Metadata of that video (before clicking on PLAY button)
Video DB
Video (Document DB)
- Video_id
- user_id
- video_type


Video DB
Video (Document DB)
- Video_id
- user_id
- video_type
- content_text
- media_url
- thumbnail_url
- like_count
-share_count
- comment_count
- other meta data
AI Based Video checking
MetaData Storage for Video
Write Ops Fast
- blocked_Videos
-filtered_video
Video against policy
Notification SVC
Moderator SVC
Pre Signed URL
KAFKA Producer
Video Consumer SVC
Filtering Videos
Video Upload Service
Check the Video's from Video DB in prior & pass them to KAFKA queue.
Video Materializer
Users_Latest_Videos
Helps in feed generation
Fanout Svc (PUSH)
< userID, List <videoID>>
KAFKA
For a specific user storing all video's to show on feed.
< video, List<subscribersUserId> >
Fanout Consumer
3 - write
3. - write
BackFill
- Read
CACHE
Feed Service
Feed Cache
Feed preparation in cache
- Read
Feed DB
1.
Will Check for the subscribers from subscriber DB
Cache
(Top People (whose video we see/ others who see our videos)
We can use AI-ML models to see the best recommendation for top models
2. Get the list of all subscribers to backfill
- Read



Outcome

- Designed a highly scalable, event-driven video streaming architecture.
- Identified critical functional and non-functional requirements for media delivery.
- Designed REST APIs and decoupled asynchronous processing pipelines.
- Understood the application of CDNs, adaptive bitrate streaming, and stream processing for massive-scale systems.